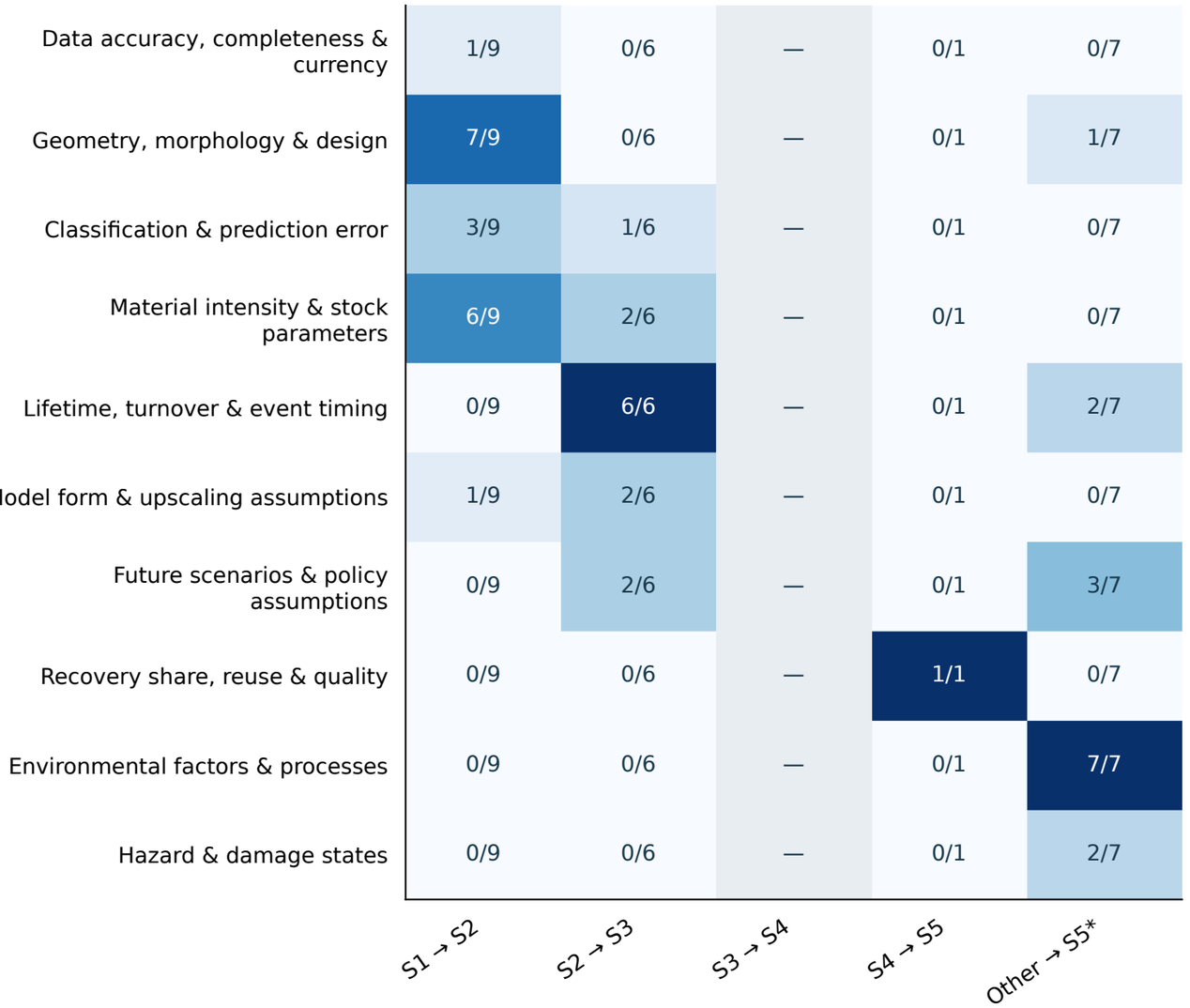
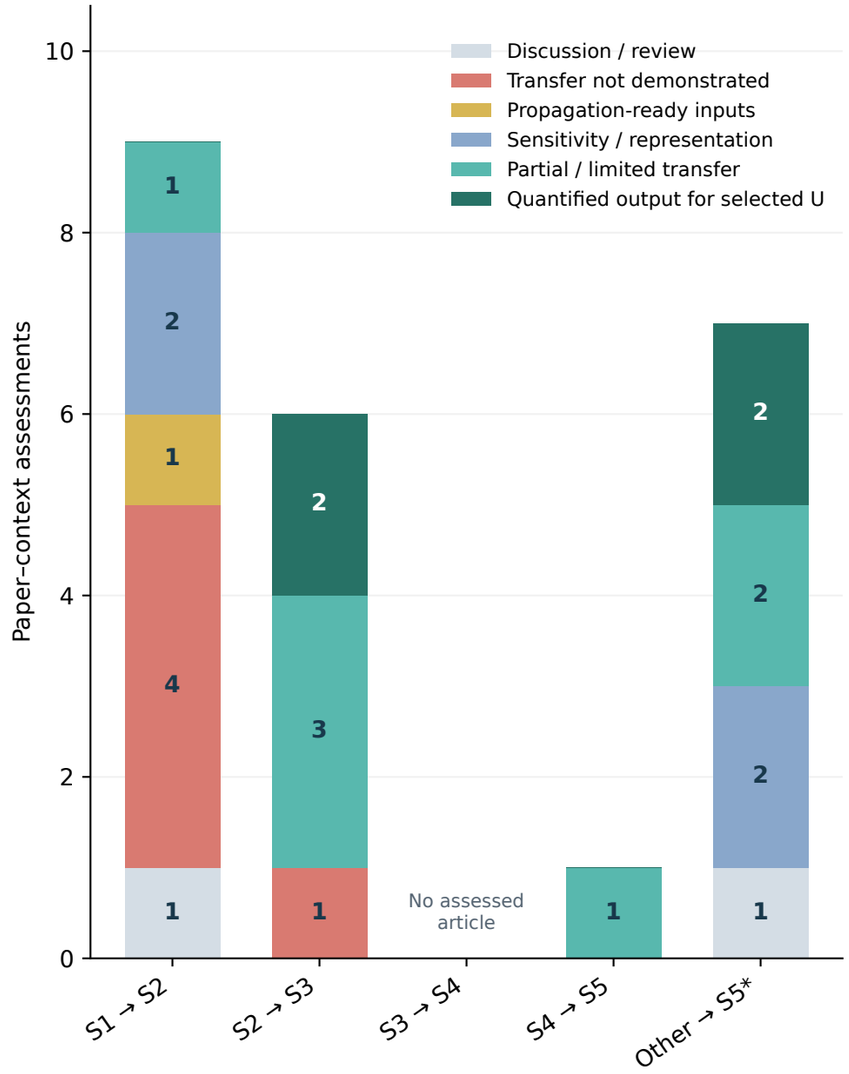
01 Stages × uncertainty

Recorded sources and continuity are different outcomes. Working set: 20 assessed papers; U10 excluded.

Source coverage
Papers / assessed context papers



Continuity evidence
One assessment per paper–context



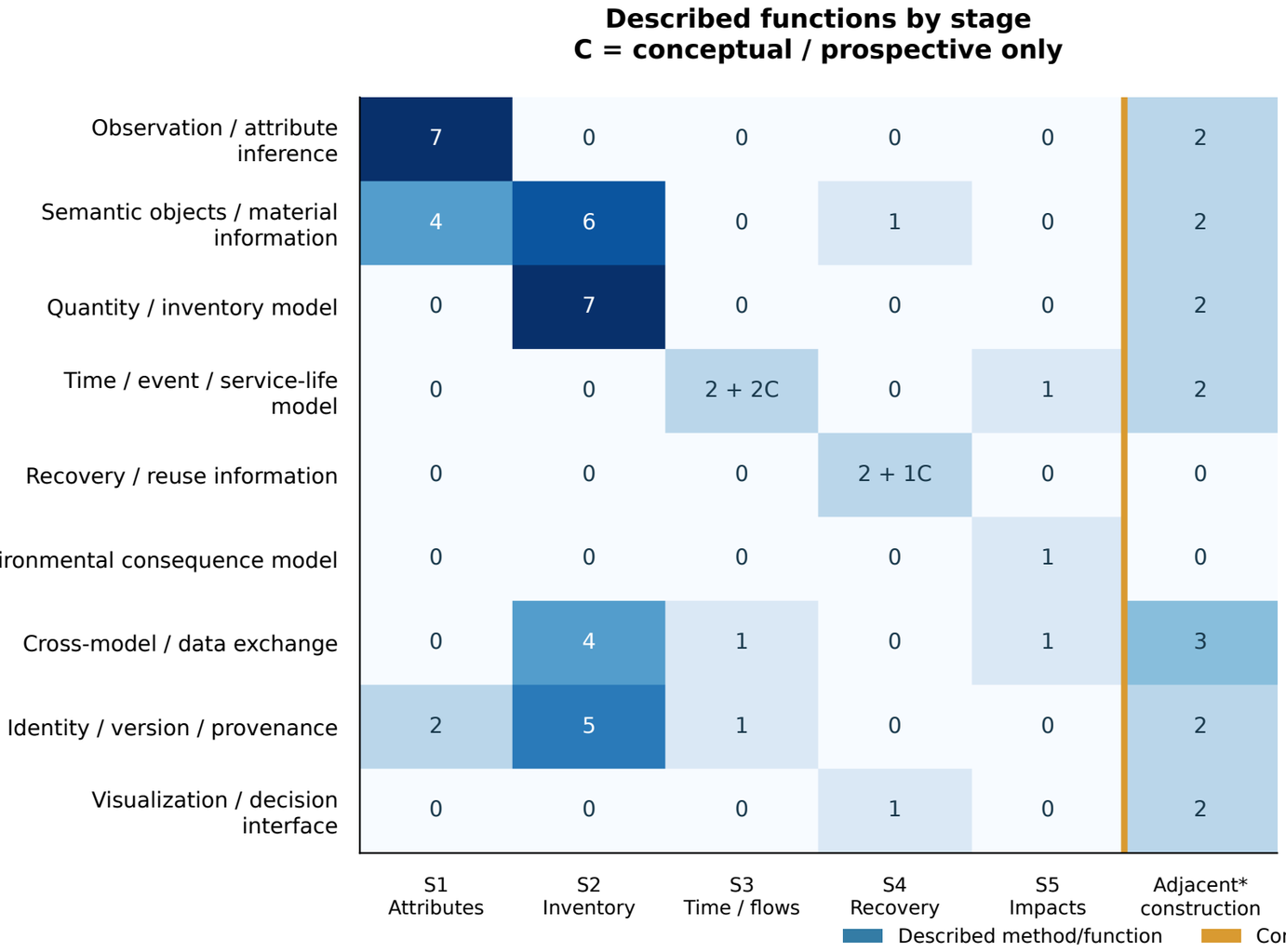
* Other → S5 keeps factor-, quantity- and damage-to-impact studies separate from recovery → impact.

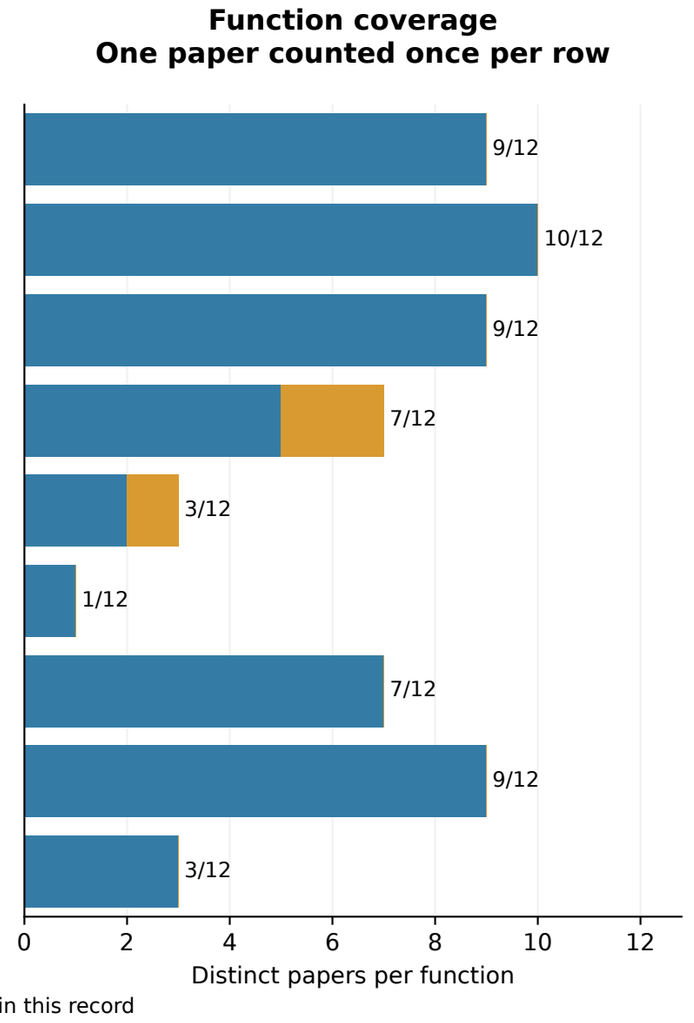
Twenty papers yield 23 context assessments. Zero means not recorded in this extraction; grey means no assessed context.

Quantified outputs for selected inputs do not establish a complete S1–S5 uncertainty chain.

02 Stages × Digital Twin and enabling functions

Twelve assessed domain / enabling studies. Functions are counted; this is not a count of twelve validated DTs.





Stage records with described functions: S1=7, S2=7, S3=2, S4=2, S5=1, X=3. Stage memberships overlap.

* Adjacent construction logistics, cost and schedule studies are not relabelled as building-stock turnover.

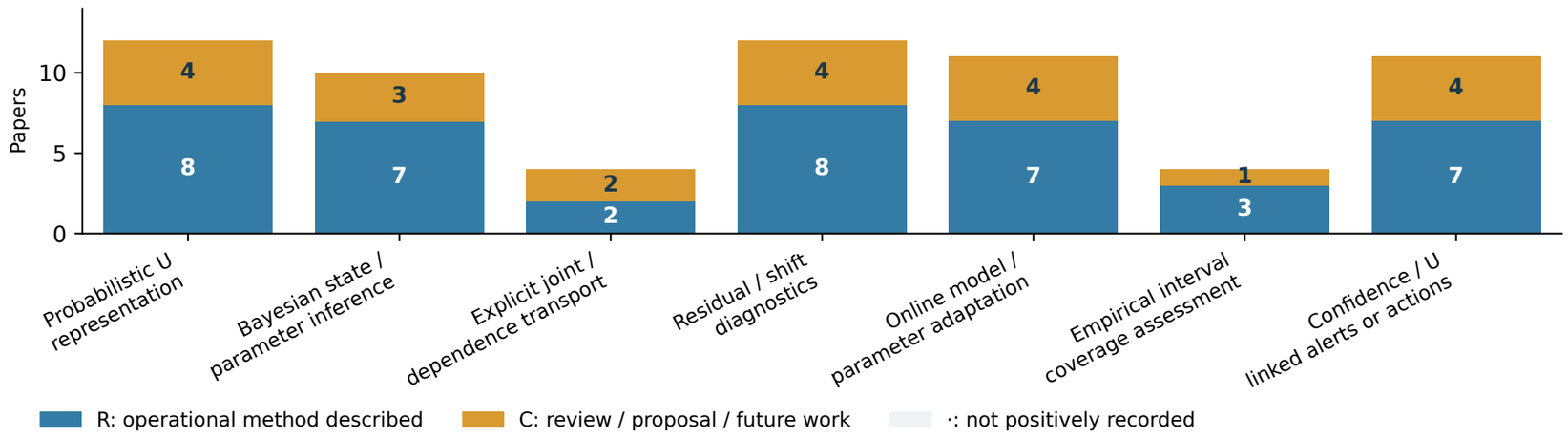
S3 includes retrospective event detection; S4 includes reuse information; neither label proves probabilistic flow/recovery modelling.

Zero = not positively coded, not proven absence. Sparse S4/S5 coverage is a property of this subset, not the entire literature.

03 Digital Twin architectures × uncertainty mechanisms

Thirteen archived appraisals. Method description, conceptual support and missing positive coding remain separate.

		Probabilistic U representation	Bayesian state / parameter inference	Explicit joint / dependence transport	Residual / shift diagnostics	Online model / parameter adaptation	Empirical interval coverage assessment	Confidence / U linked alerts or actions
Review / reference architecture	001 Physics-informed review	C	C	C	C	C	C	C
	009 Soft-sensor review	C	·	·	C	C	·	C
	010 Water-quality review	C	C	C	C	·	·	C
Hybrid model fusion	002 Hybrid fuel-system twin	R	R	·	R	R	·	R
Inference component	004 Conditional state–parameter inference	R	R	R	·	R	·	·
	008 Self-protection architecture	·	·	·	R	R	·	·
Feedback / control	015 Conflict-aware O-RAN control	R	·	·	R	R	·	R
	016 Parallel twins for NMPC	R	R	R	R	R	·	R
Layered monitoring	011 Bio360 event framework	C	C	·	R	·	·	R
	012 Field water-quality twin	R	R	·	R	C	R	R
	018 CNC tool-health twin	R	R	·	R	R	R	R
Bayesian learner	013 Hierarchical posterior evolution	R	R	·	R	R	R	R
	014 Adaptive physical-prior BNN	R	R	·	C	C	·	C

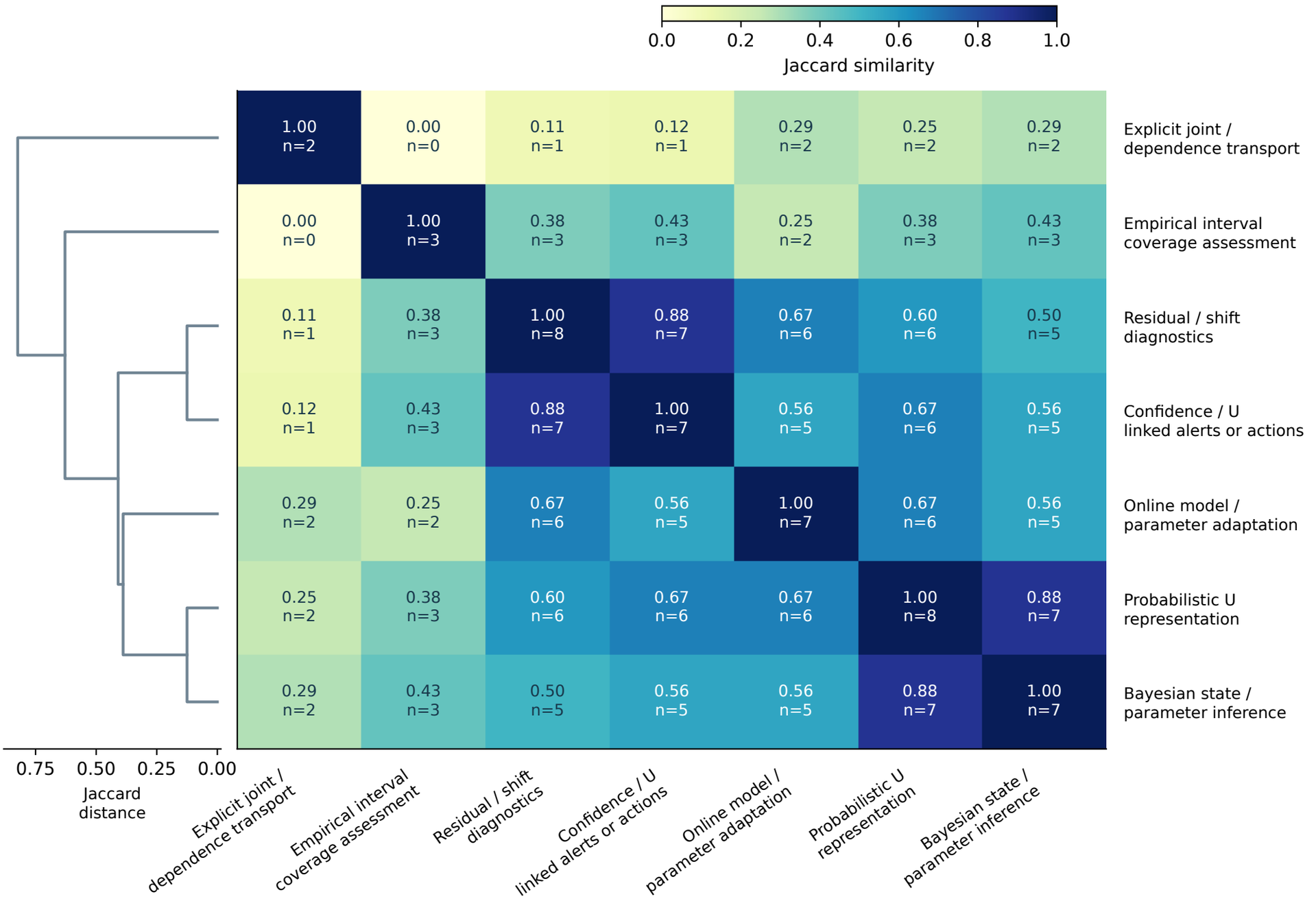


R does not mean independently validated. 004 is a preprint inference component; 015 is an adjacent control system.

Joint/dependence transport is coded for the specified variables only. It does not establish end-to-end DT continuity.

04 Which uncertainty mechanisms occur together?

Exploratory average-linkage clustering of ten method-level records. Reviews and conceptual-only tags are excluded.



Each cell shows similarity and the number of records reporting both methods. Clusters describe this selected working set.

Co-occurrence does not establish model coupling, causation or independent corroboration. Related author/study families remain a limitation.

Leave-one-record-out ranges are provided in the CSV. No significance test or stable architectural taxonomy is claimed.

05 Connecting the three evidence streams

Four boundary-specific syntheses. Existing mechanisms are not automatically validated in the material-flow domain.

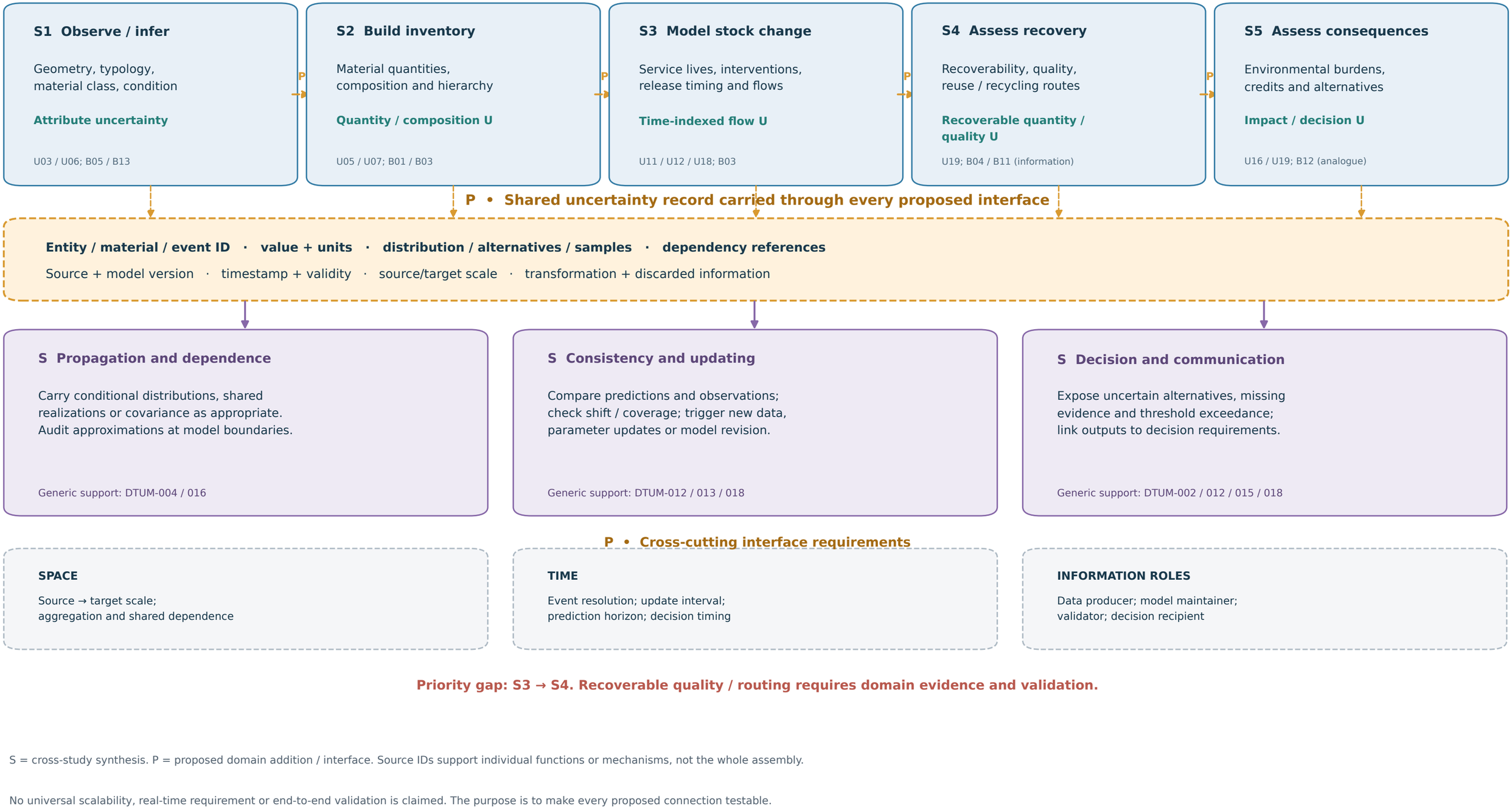
Boundary	S–U: requirement / gap	S–DT: domain function	DT–U: candidate mechanism	Synthesis / proposed addition
S1 → S2	U03 / U05 / U06 Attribute and MI variability may become point inventory.	B05 / B03 BIM–GIS quantity mapping; alternatives can be averaged.	004 / 016 Conditional or covariance representations preserve selected dependencies.	Keep material alternatives, shared dependencies and uncertainty through the quantity interface.
S2 → S3	U11 / U12 / U18 Lifetime and stock assumptions change flow timing.	B03 Layer masses + service lives produce annual outflow CSV.	004 / 016 Sequential inference and measured feedback can update represented states.	Link inventory realizations to conditioned lifetime models; revise with new event evidence.
S3 → S4	Selected-set gap No assessed direct uncertainty chain from outflow to recoverable quality.	B04 / B11 Reuse/disassembly information; not a verified probabilistic recovery-state transition.	018 (analogy) Consistency / prognostics mechanisms; own covariance handoff remains limited.	Specify condition, damage, contamination and routing interfaces. Domain data and validation are required.
S4 → S5	U19 Shared recovery inputs; joint draws and dependence insufficiently documented.	Not established here Focused domain bridge set does not establish a direct S4–S5 DT architecture.	016 (analogy) Covariance/cross-covariance transport for represented performance quantities.	Define joint recovery-to-impact input realizations and burden/credit mappings; validate in the target domain.

B aliases identify domain/enabling papers; numeric 004/016/018 labels identify DTUM studies. Full identifiers, DOIs and rationales are in the CSVs.

Dashed arrows are reviewer synthesis/proposals. Missing direct evidence is retained; no complete deployed S1–S5 system is claimed.

06 A synthesized architecture for uncertainty continuity

Material-flow functions are evidence-informed. The integrated architecture and its uncertainty-carrying interfaces are proposals for evaluation.



Priority gap: S3 → S4. Recoverable quality / routing requires domain evidence and validation.

S = cross-study synthesis. P = proposed domain addition / interface. Source IDs support individual functions or mechanisms, not the whole assembly.

No universal scalability, real-time requirement or end-to-end validation is claimed. The purpose is to make every proposed connection testable.